CSC494H1 Final Report

Development and Implementation of the Campus Eats Flutter Mobile Application

CAMPUS EATS

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INTRODUCTION

Problem: Current Solutions are Time-Consuming or Expensive

At the University of Toronto Downtown Campus, there are many food service options students can choose from, whether it be cafeterias in residence buildings, food trucks, or restaurants nearby the campus. However, many of these options require students to order in-person, which can cause heavy wait times during peak times. **Students have to spend extra time to order in-person in their already very busy schedules.** On the other hand, food delivery apps like Uber often charge high delivery and service fees, which poses limitations on students who are on a budget.

Solution: Campus Eats Mobile Application

With CampusEats mobile application, users can place orders online with ease, browse food menus, provide diverse meal options/plans, and unlock exclusive discounts tailored to students. This allows students to buy food at a price point tailored for their demographic, while also saving them time during their busy schedules.

OVERVIEW OF CONTRIBUTIONS

With any food ordering app, there are two sides; a Restaurant side and a Consumer side. Throughout this summer, I worked on both sides of the Flutter Application. My focus was in implementing the user interface design into Flutter, so some features may not be fully functional until the backend is implemented.

On the Restaurant Side, I focused on adding security within the Owner Tools tab, as well as creating more consistency with the Cancel Order workflow.

On the Consumer Side, I focused on creating new key features like Campus Location Search, Grab & Go, and Offers.

DESIGN DECISIONS AND JUSTIFICATIONS

Campus Eats Restaurant

Security: Owner Tools

Originally, users who want to access owner-specific details, like ‘View Data Analysis’, ‘Wallet’, ‘Edit Profile’, could access them through simply swiping right and opening the Side Menu. However, this poses a security risk as anyone with access to the restaurant owner account can modify core profile details. This change introduces tighter security where users must enter a password every time before they are able to access any of the Owner Tools.

In the Side Menu, I removed some of the options that fell under Owner Tools (View Data Analysis, Wallet, Edit Profile) and added an umbrella option 'Owner's Tools'. If there is a password set, clicking this will lead to a password verification screen, before accessing the Owner's Tools tab.

Within the Owner Tools Screen, I moved the existing options from Side Menu here and added an option to create and set your password. The message below all the tools changes depending on if a safety password is set. If a user presses the back button, it goes directly to the home screen.

Consistency: Cancel Order

When a restaurant owner wants to cancel an order, they are given the option to select the reason why they are canceling (like 'Kitchen Closed', 'One or More Items Unavailable', 'Other'), so as to provide information to users.

This feature was originally implemented in the homepage using a boolean functionality to alternate between showing a canceled vs. not canceled view of the order. However, this cancel functionality was not implemented everywhere. For example, the Order Detail Screen had a cancel functionality implemented but it was not consistent with the one shown in the Homepage. As it is good practice to minimize redundancy, I removed the boolean functionality and replaced it with a Cancel Order Dialog component that can be reused and have consistent functionality.

The functionality of the Cancel Order Dialog reasoning works as follows.

- Only one reason can be selected
- If no reason is selected, the confirm button is disabled. Otherwise, if a reason is selected, the confirm button is enabled.
- If ‘Other’ is selected, a text box will appear asking users to enter in their reasoning. There are no

checks yet for an empty checkbox.

When the cancel button is pressed, it goes back to the screen that was displayed before. When the confirm button is pressed, it goes to the home screen.

Campus Eats Consumer

Search Campus Location

The University of Toronto has many different campus locations, aside from the one in Downtown Toronto. Additionally, Campus Eats may want to expand to more university locations, depending on student demand.

Depending on a student's location, the restaurant and food options should also change with it. Thus, I developed a new feature that allows users to search and select their university to filter and receive better restaurant results.

When a user clicks the location icon in the top left corner of the home page, a new search screen will show up. When a user clicks on a university, the screen navigates back to the home screen, with the university location updated. When a user clicks on cancel, the screen navigates to the home screen, without any change in university location. By default, the selected university is 'University of Toronto'.

It is important to note that the list of universities is hard-coded and are just examples to demonstrate usage. In the future, we could pull this list from a database instead of hard-coding it. Further, when a user selects a new university, the restaurants shown in the homepage do not change. In the future, we can add backend logic to change restaurants based on the user's selected location.

Grab & Go

One of the key features that Campus Eats offers is Grab & Go. This feature allows users to check the food selection available to grab at a specific time period. This feature removes the barrier of having to wait for one's food, solving the issue of long wait times posed by current food options nearby campus.

This feature uses the same University Search introduced earlier to cater to each specific campus and university. We can also pick which day we want to do Grab and Go through clicking the calendar icon on the bottom right of the app bar. There is a 7 day period that users can choose from, with the first date being the current date. The functionality and implementation is fairly similar to what is found in the Edit Profile Screen when editing a user's birthday.

Further, there is a timer that counts down the difference in time between the hour of the hour box (ex. 11:00) + 20 minutes (for the grab and go period) and the current time, using a new component that calculates the difference in time. As most of the code is hard-coded, there is no ongoing functionality with the function, but a future implementation can include removing the hour box when the timer reaches 0.

Offers

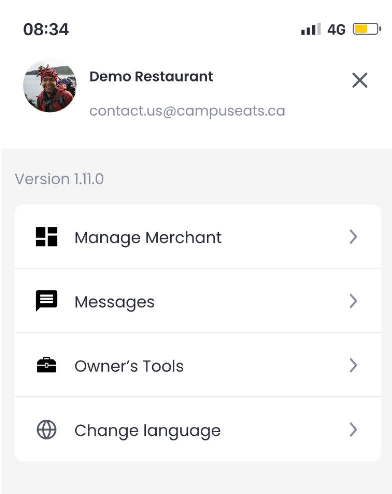
Another key feature that Campus Eats has is introducing offers exclusive to the app. These are offers that can cater to a unique student's budget. This feature removes the barrier of expensive service and delivery fees many other food delivery apps include.

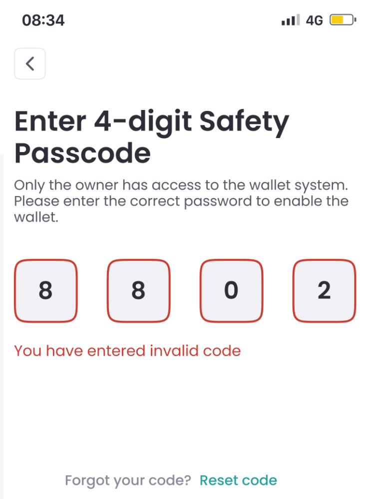
CONCLUSION

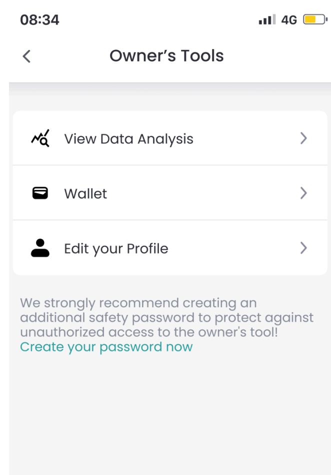
During a short period of time, I developed many key features important to both the restaurant and consumer sides of the application. On the Restaurant side, I introduced further security and consistency within the app through new workflow implementations of the Owner Tools tab and Cancel Order Reasoning. On the consumer side, I introduced many key user features, including Campus Location Search, Grab & Go, and Offers page. Through these new changes, Campus Eats identity as a convenient application for students to use is reinforced.

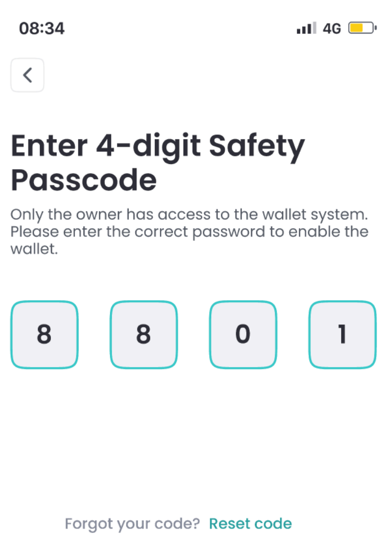
APPENDIX

Security: Owner Tools



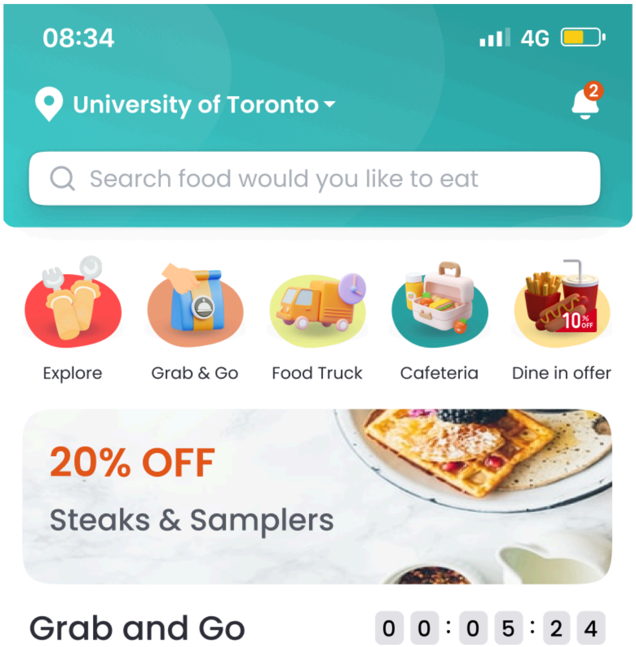


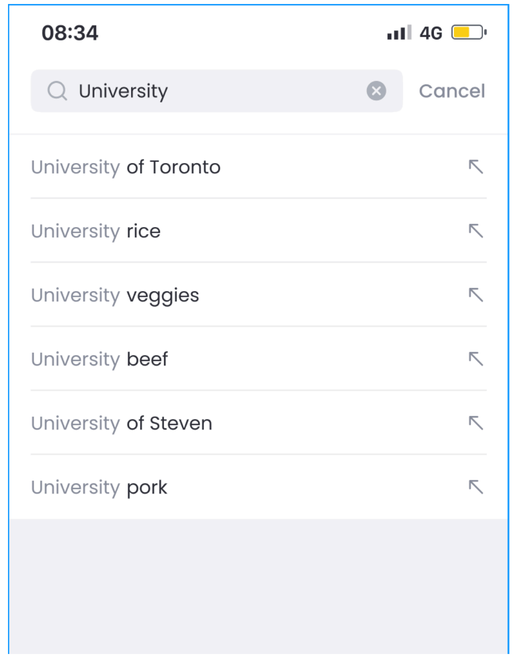




[A new UI flow]() to add an extra layer of security when users attempt to access Owner's Tools (View Data Analysis, Wallet, Edit Profile).

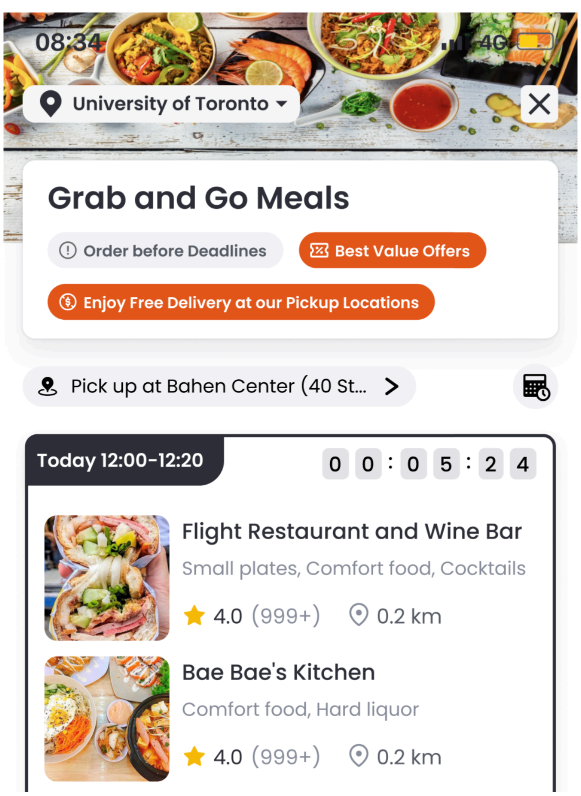
Search Campus Location

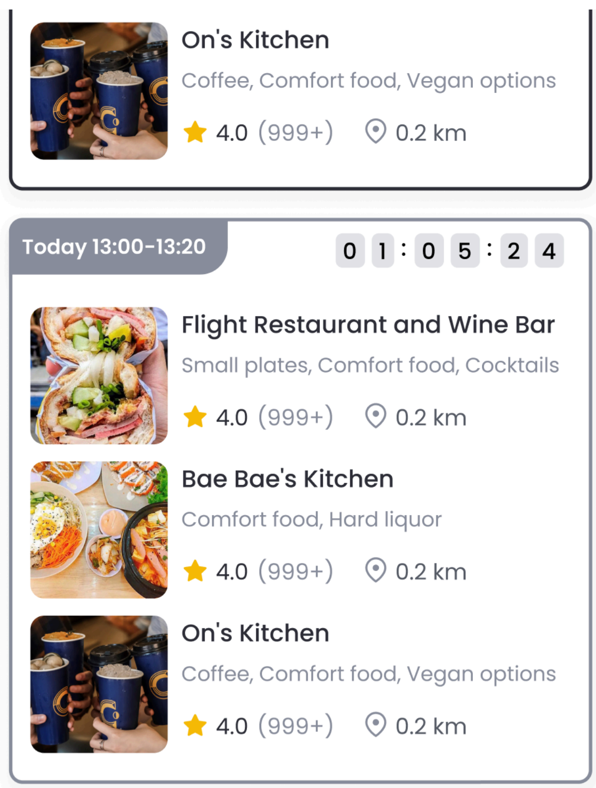




Created a new search university screen that follows [this design].

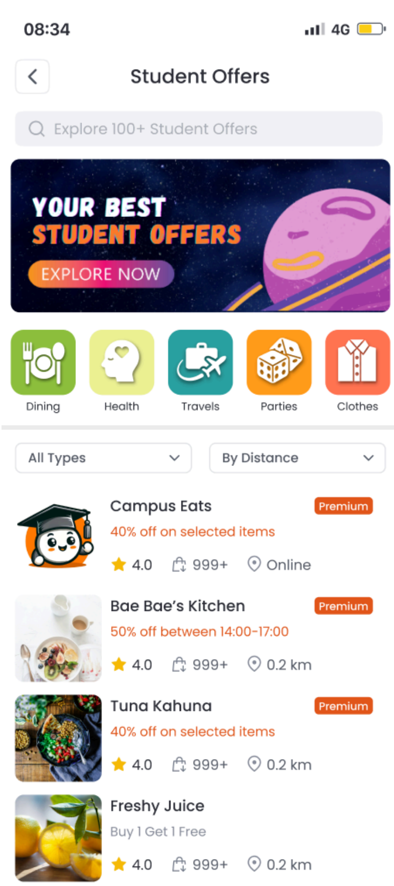
Grab & Go

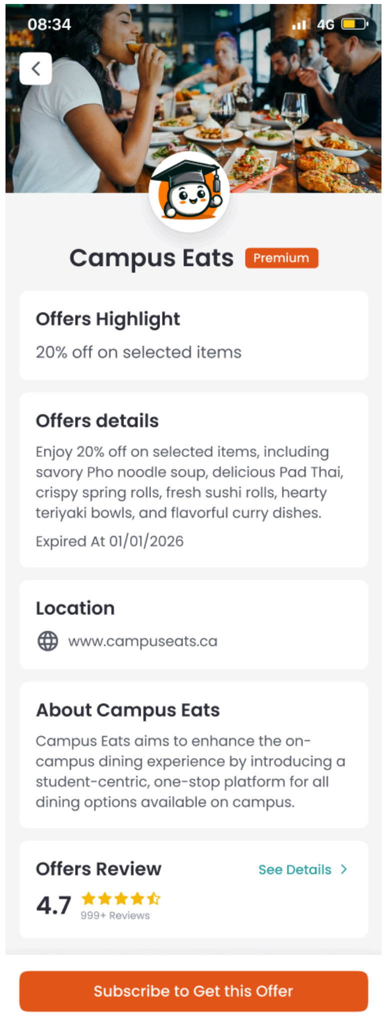




Implemented UI changes that reflect the design [here].

Offers





Implemented UI changes that reflect the design [here](#).